

9. Testing and Quality Assurance

Required by the examination paper, Part A, section 8. Test cases are recorded as **Test case** → **Expected result** → **Actual result** → **Pass/fail**, with a defect log and corrective actions in Section 9.8.

9.1 Test strategy

Testing was shaped by one architectural decision made in Phase 2: **the domain layer imports neither Flask nor SQLAlchemy**. That is what made a real test suite affordable inside the examination window, business rules can be tested with no database, no fixtures and no HTTP client, so the 186 unit tests run in **2.2 seconds**, the increase over earlier figures is Argon2 hashing in the password tests, which is deliberately slow. An Active-Record design would have required a live database for every rule test, and under a 20-hour implementation budget those tests would simply not have been written.

The levels are those set out in Session 1 [1]: unit, integration, system and acceptance. Each covers what the level below cannot:

Level	Count	Runtime	Covers	Deliberately excludes
Unit	186	2.2 s	Business rules BR-R1...R15, <code>Money</code> , service orchestration, authorisation logic, audit writes	Anything requiring SQL
Integration	42	3.3 s	Repositories, entity/record mapping, and above all the database-enforced invariants	HTTP sessions, templates
System	71	13.6 s	Routing, authentication, role authorisation, CSRF, template rendering, HTMX error path, complete user journeys	Real browsers, real devices
Total	299	20.0 s		

Why the integration level exists at all. The design claims three business invariants are enforced *twice*, in the domain layer and again by PostgreSQL partial unique indexes. A claim about PostgreSQL behaviour cannot be verified by a fake. Those tests therefore write **through the ORM, bypassing the service layer entirely**, to demonstrate that the guarantee does not depend on application code being correct.

9.2 Test environment

Runner	pytest 8 with pytest-cov
Application	Python 3.12.2, Flask 3, SQLAlchemy 2
Database	PostgreSQL 16 (Docker), separate <code>susubook_test</code> database, truncated between tests
Clock	Injected (<code>clock=lambda: date(2026, 9, 15)</code>) so date-dependent rules are deterministic
Isolation	<code>TRUNCATE ... RESTART IDENTITY CASCADE</code> per test rather than transaction rollback, because several tests deliberately provoke integrity errors that would poison an outer transaction

9.3 Coverage

app/domain/rules.py	100%	app/services/reconciliation.py	100%
app/domain/errors.py	100%	app/services/security.py	100%
app/domain/entities.py	99%	app/services/container.py	100%
app/domain/money.py	94%	app/services/collection.py	99%
app/infrastructure/models.py	100%	app/services/payout.py	98%
app/infrastructure/repositories.py	99%	app/web/collector.py	90%
app/infrastructure/qrcodes.py	100%	app/web/supervisor.py	93%
app/config.py	100%	app/web/security.py	93%
		app/web/client.py	85%
TOTAL		97%	

NFR-07 requires $\geq 70\%$ on the domain layer. Achieved: **99–100%** on domain, **97%** overall. The uncovered remainder is chiefly defensive branches — `NotImplemented` returns on `Money` arithmetic, and the `AssertionError` guard in `compute_payout` that is unreachable while the commission policy is correct.

9.4 Functional test cases

Legend: **P** pass · **F** fail. All results are from the run of the committed suite; no result is stated from memory.

TC-AUTH: Authentication (UC-01, FR-01...04)

ID	Test case	Expected	Actual	
TC-AUTH-01	Anonymous request to <code>/</code>	Redirect to <code>/login</code>	302 to <code>/login</code>	P
TC-AUTH-02	Valid collector credentials	302 to collector landing page	302 to <code>/collector/</code>	P
TC-AUTH-03	Valid client credentials	302 to <code>/my/</code>	302 to <code>/my/</code>	P
TC-AUTH-04	Valid supervisor credentials	302 to <code>/supervisor/</code>	302 to <code>/supervisor/</code>	P
TC-AUTH-05	Correct phone, wrong password	401, generic message	401, "Phone number or password is incorrect"	P
TC-AUTH-06	Unknown phone number	Identical status and message to TC-AUTH-05	Both 401, both same message	P
TC-AUTH-07	Logout	Session cleared, protected page redirects	302 on subsequent <code>/collector/</code>	P
TC-AUTH-08	Failed login is audited	<code>LOGIN_FAILED</code> row written	Row present with reason	P

TC-AUTH-06 is a security test in functional clothing: distinguishing an unknown phone from a wrong password would let an attacker enumerate which numbers are registered.

TC-AUTHZ: Authorisation (FR-05, NFR-03, BR-R15)

ID	Test case	Expected	Actual	
TC-AUTHZ-01	Unauthenticated access to /collector/ , /supervisor/ , /my/	All redirect	All 302	P
TC-AUTHZ-02	Collector opens supervisor pages	403	403	P
TC-AUTHZ-03	Client opens collector pages	403	403	P
TC-AUTHZ-04	Collector opens a client self-service page	403	403	P
TC-AUTHZ-05	Collector B opens Collector A's client via QR reference	403, "not on your route"	403, message present	P
TC-AUTHZ-06	Collector B posts a collection for that client	403, nothing written	403, no contribution row	P
TC-AUTHZ-07	Denied attempt is audited	AUTHORISATION_DENIED row	Row present	P (after DEF-05)
TC-AUTHZ-08	Collector B prints that client's QR card	403	403	P

TC-AUTHZ-05 through 08 are the photographed-card threat, and the reason BR-R15 exists. A QR card carried through a market must be assumed photographable; these cases demonstrate that possession of the reference confers no capability, because the authorisation decision never consults it.

TC-COL: Recording a contribution (UC-03)

ID	Test case	Expected	Actual	
TC-COL-01	Route sheet lists only own clients	Other collectors' clients absent	Absent	P
TC-COL-02	Scan landing page	Pre-filled daily rate, confirm button	"GHS 10.00", "Confirm collection"	P
TC-COL-03	Record a contribution	302, one row, reference assigned	302, SB- reference present	P
TC-COL-04	Second contribution, same client, same day (BR-R5)	422, existing reference quoted	422, "already recorded (reference SB-...)"	P
TC-COL-05	Duplicate over HTMX	422 fragment , not a full page	Fragment, no <code><html></code>	P
TC-COL-06	Amount not a multiple of the rate (BR-R7)	422	422, "whole multiple"	P
TC-COL-07	Unparseable amount ("abc")	422	422	P
TC-COL-08	Unknown client reference	404	404	P
TC-COL-09	Future-dated contribution (BR-R4)	Refused	<code>ContributionDateInFuture</code> raised	P
TC-COL-10	Contribution outside cycle dates (BR-R3)	Refused	<code>ContributionDateOutsideCycle</code> raised	P
TC-COL-11	Contribution into a closed cycle (BR-R6)	Refused	<code>CycleClosed</code> raised	P
TC-COL-12	HTMX collect updates the running total out of band (DEF-06)	Response carries the day's total marked for out-of-band swap, with the new value	<code>id="recorded-today"</code> , <code>hx-swap-oob="true"</code> , "Recorded GHS 10.00"	P
TC-COL-13	Route sheet total after a collection	GHS 0.00 before, GHS 10.00 after	As expected	P

TC-ENR: Enrolment (UC-02)

ID	Test case	Expected	Actual	
TC-ENR-01	Enrol a client	Client, login and cycle 1 created atomically	All three present, cycle ACTIVE	P
TC-ENR-02	New client signs in immediately	302 to /my/	302 to /my/	P
TC-ENR-03	Password shorter than 6 characters	422	422	P
TC-ENR-04	Zero daily rate	Refused	"must be more than zero"	P
TC-ENR-05	Client receives an opaque public reference	UUIDv4, not the row id	36-char UUID, differs from id	P
TC-ENR-06	Cycle snapshots the daily rate	Cycle rate equals agreed rate	Equal	P

TC-PAY: Payout (UC-07)

ID	Test case	Expected	Actual	
TC-PAY-01	Matured cycle appears for release	Client listed	Listed	P
TC-PAY-02	Release computes commission (BR-R8)	10 days $\times$ GHS 10 $\rightarrow$ total 100, commission 10, net 90	10 000 / 1 000 / 9 000 pesewas	P
TC-PAY-03	Money is conserved	net + commission = total	Asserted equal	P
TC-PAY-04	Cycle closes and the next opens	[PAID_OUT, ACTIVE]	[PAID_OUT, ACTIVE]	P
TC-PAY-05	New cycle inherits the rate	Same daily rate	Equal	P
TC-PAY-06	One-day client (BR-R9)	Net exactly zero, never negative	total 1 000, commission 1 000, net 0	P
TC-PAY-07	Client who paid nothing	Zero throughout, no negative	All zero	P
TC-PAY-08	Payout never negative, all 32 completion levels	Non-negative and conserved for 0...31 days	All 32 pass	P
TC-PAY-09	Second release (BR-R10)	422	422, "already been paid out"	P
TC-PAY-10	Release before maturity (BR-R12)	Refused	CycleNotMatured raised	P
TC-PAY-11	Collector attempts release	403	403	P
TC-PAY-12	Release is audited	RELEASE_PAY0UT with settlement	Row with all three amounts	P

TC-PAY-08 is exhaustive rather than sampled. BR-R9's edge case, a client whose entire balance is consumed by commission, is exactly where an off-by-one produces a *negative payout*, so every completion level from 0 to 31 days is checked for non-negativity and conservation.

TC-REC: Reconciliation (UC-05, UC-06, FR-24...26)

ID	Test case	Expected	Actual	
TC-REC-01	Declared cash matches recorded	Variance zero, reconciled	GHS 0.00, reconciled	P
TC-REC-02	Declared less than recorded	Shortfall shown to supervisor	GHS 7.00 shortfall, "unreconciled"	P
TC-REC-03	Collector never declared	Flagged as not declared	"NOT DECLARED"	P
TC-REC-04	Collector recorded nothing	Still listed, zeroes	Listed with zeroes	P
TC-REC-05	Negative declaration	Refused	"cannot be negative"	P
TC-REC-06	Future-dated declaration	Refused	"future date"	P
TC-REC-07	Collector views all variances	403	403	P
TC-REC-08	Re-declaring the same day overwrites	Latest value stored	GHS 25.00 replaces GHS 10.00	P
TC-REC-09	Declaration is audited with the variance	Variance in audit detail	700 pesewas recorded	P

TC-REC-04 matters more than it looks: a collector who simply stopped working must still appear on the variance screen. If idle collectors dropped off the report, the most suspicious case would be the one that became invisible.

TC-REV: Reversal (UC-09, BR-R11)

ID	Test case	Expected	Actual	
TC-REV-01	Supervisor reverses a contribution	Original preserved and linked	2 rows, <code>reversed_by_id</code> set	P
TC-REV-02	Reversal frees the day for a replacement	New contribution accepted	302, accepted	P
TC-REV-03	Reversing twice	Refused	"already been reversed"	P
TC-REV-04	Reason is required	422	422	P
TC-REV-05	Collector attempts reversal	403	403	P
TC-REV-06	Reversal is audited with the reason	Reason in audit detail	Present	P
TC-REV-07	Client still sees both entries	REVERSED and REVERSAL both visible	Both rendered	P

TC-CLI: Client self-service (UC-08, FR-28...30)

ID	Test case	Expected	Actual	
TC-CLI-01	Client sees their own contributions	Card renders with entries	200, entries present	P
TC-CLI-02	Each entry names the recording collector	Collector name shown	"Joseph Osei" present	P
TC-CLI-03	Balance and projected payout shown	"You will receive"	Present	P
TC-CLI-04	Card renders every day of the cycle	31 boxes	31 <code>role="listitem"</code>	P
TC-CLI-05	Past cycles page	200	200	P
TC-CLI-06	Reversed entries remain visible to the client	Both labels shown	REVERSED and REVERSAL present	P

TC-CLI-02 is the single most important functional test in this document. It is the answer to problem **P1**: the client's record is independent of the collector, and every entry is attributable.

TC-QR: QR client cards (FR-39, FR-40, CR-001)

ID	Test case	Expected	Actual	
TC-QR-01	Card page renders inline SVG	<svg> present	Present	P
TC-QR-02	Card discloses no phone number	Phone absent from page	Absent	P
TC-QR-03	Another collector cannot print the card	403	403	P

TC-PWD: Credential management (FR-41, FR-42; TD-15 repayment)

ID	Test case	Expected	Actual	
TC-PWD-01	Flagged user is redirected from every page	All redirect to /password	/my/ , /my/history , /all 302	P
TC-PWD-02	The change page itself remains reachable	200	200	P
TC-PWD-03	Logout permitted while blocked	302	302	P
TC-PWD-04	Changing lifts the block	Record accessible afterwards	200	P
TC-PWD-05	The collector's password stops working	Old refused, new accepted	401 then 302	P
TC-PWD-06	Unflagged user is not obstructed	200	200	P
TC-PWD-07	Forced change does not demand the old password	Accepted	Accepted	P
TC-PWD-08	Voluntary change requires the current password	Refused	"not correct"	P
TC-PWD-09	Weak password refused	422	422	P
TC-PWD-10	Reset code stored only as a hash	Plaintext absent	\$argon2... , plaintext absent	P
TC-PWD-11	Unknown number gives an identical response	Indistinguishable	Same status and message	P
TC-PWD-12	End-to-end reset by code	New password works	302 on subsequent login	P
TC-PWD-13	Code single-use, expiring, attempt-capped, rate-limited	All four refuse	All refuse	P

TC-PWD-05 is the test that matters. TD-15 was Critical because the collector knew the client's password, making the client's record dependent on them and contradicting BR-02. This case asserts the collector's password no longer works once the client has chosen their own.

TC-SMS: Client notification (FR-31, CR-002)

The negative cases matter most here. The demonstration dataset uses valid-format Ghanaian numbers that may belong to real people, so an unguarded send would text strangers on every recorded collection.

ID	Test case	Expected	Actual	
TC-SMS-01	Default configuration sends nothing	No message, no <code>SMS_DISPATCHED</code> audit	Gateway untouched, audit absent	P
TC-SMS-02	Every seeded phone number is refused by default	All 15 refused	All 15 refused	P
TC-SMS-03	Allowlisted client is notified	Message dispatched to the normalised number	<code>233201000202</code> , correct body	P
TC-SMS-04	Dispatch is audited	<code>SMS_DISPATCHED</code> written	Present	P
TC-SMS-05	A failing gateway does not fail the collection	302, contribution still committed	302, one contribution row	P
TC-SMS-06	Number forms normalise identically	<code>024...</code> , <code>+233...</code> , <code>233...</code> , spaced all match	All match	P
TC-SMS-07	Unparseable number refused even under allow-all	Refused	Refused	P
TC-SMS-08	Message fits one SMS segment	≤160 characters	Within limit	P
TC-SMS-09	Message carries no link, credential or token	None present	None present	P
TC-SMS-10	Message carries amount, date, collector, reference, total	All five present	All five present	P

TC-SMS-02 is a regression guard against a specific harm rather than a rule. It asserts that every number in the seed dataset is refused by a default-configured service, so the safety gate cannot be removed without a test failing.

TC-DB: Database-enforced invariants (defence in depth)

Written through the ORM, **bypassing the service layer**, so the guarantee is shown not to depend on application code.

ID	Test case	Expected	Actual	
TC-DB-01	Second ACTIVE cycle for one client (BR-R2)	Rejected	ux_active_cycle_per_client violation	P
TC-DB-02	A new cycle after the first is paid out	Permitted	Committed	P
TC-DB-03	Two clients each with an active cycle	Permitted	Committed	P
TC-DB-04	Duplicate contribution on one day (BR-R5)	Rejected	ux_effective_contribution_per_day violation	P
TC-DB-05	Replacement shares a date with a reversed entry	Permitted; all three rows survive	Committed, 3 rows, 1 effective	P
TC-DB-06	Duplicate contribution reference	Rejected	Unique violation	P
TC-DB-07	Second payout for one cycle (BR-R10)	Rejected	Unique violation	P
TC-DB-08	Payout where net + commission $\neq$ total	Rejected	ck_payout_balances violation	P
TC-DB-09	Negative net payout	Rejected	Check violation	P
TC-DB-10	Zero or negative daily rate	Rejected	ck_client_rate_positive	P
TC-DB-11	Zero contribution amount	Rejected	ck_contribution_positive	P
TC-DB-12	Cycle end date before start	Rejected	ck_cycle_dates	P
TC-DB-13	Unknown user role	Rejected	ck_user_role	P
TC-DB-14	Two remittance declarations, one collector, one day	Rejected	uq_declaration_per_day	P

TC-DB-05 is the test that protects the design. The index on contributions is *partial* precisely so a reversed entry, its reversal, and the replacement may share a date. A future developer "simplifying" it to a plain `UNIQUE(cycle_id, contribution_date)` would make correction by reversal impossible, and this test is what would stop them.

TC-MON: Money handling (BR-R1, NFR-04)

ID	Test case	Expected	Actual	
TC-MON-01	Construct from "2.50"	250 pesewas	250	P
TC-MON-02	Construct from float 2.50	TypeError	Raised	P
TC-MON-03	Multiply by float	TypeError	Raised	P
TC-MON-04	Money(True), bool is an int subclass	TypeError	Raised	P
TC-MON-05	Fractional pesewa ("2.505")	ValueError	Raised	P
TC-MON-06	Exact accumulation across all cycle lengths 1–31	Exact at every length	All exact	P
TC-MON-07	Money survives the ORM mapping	1234 pesewas in and out	Exact, still int	P

9.5 Security testing (NFR-03)

ID	Test case	Expected	Actual	
TC-SEC-01	POST without a CSRF token, enforcement on	Rejected	400	P
TC-SEC-02	Password stored hashed	Argon2id, plaintext absent	\$argon2..., plaintext absent	P
TC-SEC-03	Password never rendered in a page	Absent from response	Absent	P
TC-SEC-04	URLs expose opaque references, not row ids (BR-R14)	UUID in links, no /client/1	UUID present, sequential absent	P
TC-SEC-05	Account enumeration via login response	Indistinguishable	Same status and message	P
TC-SEC-06	Forced browsing to another role's pages	403	403	P
TC-SEC-07	Unknown route	404, no stack trace	404	P

Security findings not covered by a passing test, carried into the debt register rather than presented as satisfied:

- **TD-14**, no rate limiting or lockout. Brute force is possible; failed attempts are audited, so an attack is visible afterwards but nothing stops it. Classified **Critical**.
- **TD-15**, the collector sets the client's initial password and there is no forced change, so the collector can sign in as the client. Classified **Critical**, because it undermines the independence the system exists to provide.
- **TD-09**, the audit log is append-only by application convention, not by database permission. Classified **Critical**.

No penetration testing or automated dependency vulnerability scanning was performed; both are named as gaps in Section 9.10.

9.6 Performance testing (NFR-01)

Server-side render time, median of five requests against local PostgreSQL with seeded data (10 clients, ~170 contributions):

Page	Server time	HTML size
Collector route sheet	9 ms	7.5 KB
Enrolment form	2 ms	4.3 KB
Supervisor variances	3 ms	2.3 KB
Client susu card (31 boxes + full history)	12 ms	28.6 KB

Interpretation. These figures measure *server* time only. NFR-01 specifies 2 seconds end-to-end on a 3G-class connection, which server time does not establish: the dominant cost on that connection is network transfer plus the render-blocking Tailwind CDN request (**TD-02**), neither of which is measured here. The figures indicate that the application's own work leaves the entire budget available for transport, and that **TD-02 is currently the largest threat to NFR-01**, which is why it is scheduled for repayment.

The 28.6 KB client card is the largest page and grows with contribution count; the route sheet's N+1 (**TD-16**) grows with route size. Neither is a problem at demonstration scale and both are recorded.

Not performed: load testing, concurrency testing, testing against a real 3G connection or a real low-end Android device. Named as gaps in Section 9.10.

9.7 Requirements verification

Requirement	Status	Evidence
FR-01...05 authentication and authorisation	✓ Satisfied	TC-AUTH, TC-AUTHZ
FR-06, FR-07 enrolment	✓ Satisfied	TC-ENR
FR-08 client list and search	⚠ Partial	List renders; search and pagination not implemented (TD-03)
FR-10...14 cycles and validation	✓ Satisfied	TC-COL-03...11, TC-DB-01...06
FR-16, FR-17 susu card and summary	✓ Satisfied	TC-CLI-04, unit summary tests
FR-18...21 payout	✓ Satisfied	TC-PAY
FR-23 route sheet	⚠ Partial	Renders with status; not filtered or route-ordered (TD-05)
FR-24...26 reconciliation	✓ Satisfied	TC-REC
FR-28...30 client transparency	✓ Satisfied	TC-CLI
FR-32, FR-33 audit and reversal	✓ Satisfied	TC-REV, audit assertions
FR-39, FR-40 QR cards	✓ Satisfied	TC-QR
FR-31 SMS notification	⚠ Partial	TC-SMS passes; restricted to an allowlist and never verified on a real handset
FR-41 forced password change	✓ Satisfied	TC-PWD-01...09
FR-42 password reset	⚠ Partial	TC-PWD-10...13 pass; delivery depends on FR-31, so a client outside the allowlist cannot self-serve
NFR-01 performance	⚠ Partial	Server time measured; end-to-end on 3G not measured
NFR-02 usability (≤ 3 interactions)	⚠ Unverified	Design achieves 2 by inspection; not measured with a participant
NFR-03 security	⚠ Partial	TC-SEC passes; TD-09, TD-14, TD-15 outstanding
NFR-04 money integrity	✓ Satisfied	TC-MON, TC-DB-08
NFR-06 data minimisation	✓ Satisfied	Schema review, TC-QR-02
NFR-07 maintainability ($\geq 70\%$ domain)	✓ Satisfied	99–100% domain coverage
NFR-08 accessibility	⚠ Partial	Shape+text encoding and 44px targets implemented; no contrast audit or screen-reader test
NFR-09 auditability	✓ Satisfied	Audit assertions across all suites
NFR-10 portability	✓ Satisfied	Same engine dev and prod; deploy verification pending Phase 5

Three requirements are **partially** satisfied and three NFRs are **unverified or partial**. They are reported as such rather than claimed complete.

9.8 Defect log

Defects found during development, with corrective action. All are closed.

ID	Defect	Found by	Severity	Corrective action	Status
DEF-01	A unit test and a module docstring both asserted that $31 \times \text{GHS } 0.10$ accumulates to 3.0000000000000004 in floating point. It does not, that sum rounds back to exactly 3.1. The example did not demonstrate the error it claimed.	Unit suite (the test failed)	Low (documentation), High (reasoning)	Verified empirically; corrected to $n=29$ (2.9000000000000004). Test rewritten to assert exactness across the whole 1–31 range rather than at a single sample, and to record that float error here is <i>intermittent</i> , which is worse than consistently wrong, because it survives casual testing.	Closed
DEF-02	Boolean and status columns carried Python-side defaults only. A direct SQL insert omitting them failed on NOT NULL <i>before</i> the partial unique index was consulted.	Manual SQL probing of the invariants	Medium	Added <code>server_default</code> . Surviving direct writes is the entire purpose of those indexes, so a schema that only works when written through the ORM defeated the design.	Closed
DEF-03	<code>Flask(__name__)</code> looked for templates at the package root; every page returned 500.	Smoke test after first run	High	Pointed <code>template_folder</code> at <code>app/web/templates</code> to match the layering.	Closed
DEF-04	<code>cycle_days</code> was registered as a context processor, but Jinja macros do not receive the template context. Every susu card view raised <code>UndefinedError</code> , the client card, the collector's client detail, both broken.	Manual page walk	High	Moved to <code>app.jinja_env.globals</code> .	Closed
DEF-05	<code>_assert_on_route</code> in the web layer raised <code>NotAuthorised</code> without auditing , while the service-layer path did audit. A collector presenting a client reference they should not hold was recorded only if they attempted a <i>write</i> ; a denied GET on a scanned card left no trace.	System suite (TC-AUTHZ-07)	Medium–High	Audit on both paths. The denial arrives on the GET, before any write is attempted, and is exactly the signal a supervisor needs.	Closed
DEF-06	On the route sheet, recording a contribution over HTMX swapped the client's row to "Paid" but left the day's running total unchanged until a manual	Manual exploratory testing by the project owner	Medium	<code>hx-target</code> swapped only the row; the total sat outside it. Returned the total in the same response as an out-of-band swap (<code>hx-swap-oob</code>), so one request updates both. Two regression tests added.	Closed

ID	Defect	Found by	Severity	Corrective action	Status
	refresh. The row and the total disagreed on the same screen.				
DEF-07	The security test asserting that URLs expose opaque references, not sequential ids, was flaky : it passed alone and failed roughly one full-suite run in sixteen.	Full-suite run after fixing DEF-06	Medium	The negative assertion used substring matching — <code>/collector/client/1</code> is a substring of <code>/collector/client/1a2b3c...</code> , so it failed whenever a random UUID began with the digit 1. Rewritten to match a complete numeric path segment by regex. Verified stable over five consecutive full runs.	Closed

Two of the developer's own test assumptions were also incorrect. Both were corrected rather than accommodated by weakening the assertion until it passed:

	Wrong assumption	Reality	Resolution
1	Setting <code>status = 'MATURED'</code> makes a cycle due for payout	<code>list_due_for_payout</code> filters on <code>end_date < today</code> , which is what BR-R12 actually describes	Tests now move the cycle genuinely into the past
2	<code>list_for_target(type, None)</code> returns nothing, since <code>= NULL</code> never matches	SQLAlchemy renders <code>column == None</code> as <code>IS NULL</code> , so it does match	Test rewritten to assert what it actually claimed, that an anonymous actor can be stored

Defect density: 7 defects across roughly 1,090 statements. Five (DEF-01, DEF-02, DEF-05, DEF-07, and the two corrected assumptions) were found by tests or by probing rather than by a user, which is the outcome the test strategy was designed to produce.

DEF-06 is the exception. It was identified by manual interaction with the interface rather than by any automated test, and it could not have been found by them, because every test asserted the *response* to a collect request and none asserted the *state of the page afterwards*. The suite verified that the correct row was returned; the inconsistency in the total above it was observable only to a person. This provides direct evidence for the value of the user acceptance testing recorded as outstanding at Section 9.9: automated tests confirm the behaviour the developer thought to specify, whereas a user encounters behaviour the developer did not anticipate.

DEF-07 illustrates the opposite risk. An intermittently failing test is more damaging than no test, because it conditions a team to disregard a failing result, and this one guarded a security property (BR-R14). It was fixed rather than retried, and stability was then demonstrated over five consecutive full runs rather than assumed.

9.9 User acceptance testing: status

Not yet conducted with an independent participant. This is stated plainly rather than substituted for.

The 51 system tests exercise complete user journeys end to end, but they were written by the developer and are therefore **system tests, not user acceptance tests**. Presenting them as UAT would misrepresent what they establish.

The prepared UAT protocol, for execution with a participant:

ID	Scenario	Role	Acceptance criterion
UAT-01	Sign in and record a day's collection for three clients	Collector	Completed without assistance; each recorded in ≤ 3 interactions
UAT-02	Attempt to collect twice from one client on the same day	Collector	Refusal understood without explanation
UAT-03	Enrol a new client and print their QR card	Collector	Client, login and card produced; card scans
UAT-04	Declare the day's cash, under-declaring deliberately	Collector	Variance understood as a discrepancy
UAT-05	Sign in and confirm every payment made this week	Client	Locates own record; confirms amounts and collector names
UAT-06	State what will be received at maturity and why the deduction exists	Client	Identifies payout and explains the one-day commission
UAT-07	Identify which collector is unreconciled today	Supervisor	Locates the variance without prompting
UAT-08	Release a matured payout	Supervisor	Understands the settlement before confirming
UAT-09	Reverse a wrongly recorded contribution	Supervisor	Confirms the original remains visible

Usability measures to be captured alongside (Session 5 [5]): task completion rate, time on task, error rate, and a 10-item System Usability Scale score [23]. The nine-scenario protocol uses a small participant group on Nielsen's finding that five users surface the large majority of usability problems [25]. NFR-02's claim that a routine collection takes ≤ 3 interactions is currently established **by design inspection only**, scan plus confirm is two, and needs a measured value from a real participant to be reported as verified.

9.10 Known gaps in testing

Recorded so the coverage claim is not overstated:

Gap	Consequence	Why
No UAT with an independent participant	NFR-02 unverified; usability findings unknown	No participant available within the window (Section 9.9)
No real-device or real-network testing	NFR-01 unverified end-to-end; NFR-08 outdoor legibility unverified	No 3G connection or low-end Android available
No load or concurrency testing	Behaviour under simultaneous collectors unknown	Out of scope for the window
No SMS delivered to a real handset	FR-31 verified in mechanism only; the gateway's actual behaviour is untested	Recipients are restricted to an allowlist, and no allowlisted handset was available
No accessibility audit or screen-reader test	NFR-08 partially verified	Requires tooling and time not budgeted
No penetration testing or dependency vulnerability scan	Unknown vulnerabilities may exist	Out of scope; TD-12 also leaves dependencies unpinned
No browser-matrix testing	Rendering verified in Chromium only	Single-browser check only
QR scanning not tested on a physical printed card	FR-40 verified as a URL route, not as an optical read	No printer or camera test performed

The final gap is significant. **TC-QR verifies that the card renders and that the route resolves and authorises correctly, but no card has been printed and read optically with a handset.** That step belongs in UAT-03.